RBCET: A Reputation-Based Blockchain Consensus Mechanism for Fast and Secured Energy Trading in Internet of Electric Vehicles (IoEV)

Amritesh Kumar, *Member, IEEE*, Koustav Kumar Mondal, *Graduate Student Member, IEEE*, Lokendra Vishwakarma, *Member, IEEE*, and Debasis Das, *Senior Member, IEEE*

***Abstract*—The state-of-the-art consensus mechanisms in blockchain-based Internet of Electric Vehicles (IoEV) prove suboptimal for energy trading due to their heavy CPU workload, high bandwidth consumption, energy inefficiencies, and the backlog of pending transactions. Also, the network is susceptible to attacks such as double-spending, Distributed Denial of Service (DDoS), eclipse, Sybil, and collusion attacks. Addressing these challenges, this paper introduces RBCET, a reputation-based consensus mechanism explicitly designed for energy trading within the IoEV blockchain network. The RBCET computes the reputation score (RS) of nodes (e.g., Electric Vehicles (EVs) and Charging Stations (CSs)) in the IoEV network using a decision tree regressor (DTR) model. This model utilizes the reputation parameter dataset generated by a relational mathematical model. RBCET allows only those EVs and CSs to participate in energy trading whose RS value exceeds the average RS value. Additionally, we also propose a hardware-based incentive mechanism to calculate and store incentives to enhance the efficiency of energy trading. Security analysis shows that RBCET resists the attacks mentioned above. Furthermore, we have developed a testbed using 10 Raspberry Pi 4 devices to demonstrate the performance of RBCET compared with state-of-the-art schemes. This comparison shows that RBCET reduces CPU workload by 47%, energy consumption by 10%, bandwidth usage by 5%, the volume of pending transactions by 75%, and overall improves efficiency by 62%.**

***Index Terms*—Blockchain, consensus algorithm, electrical vehicle, energy trading, Internet of electric vehicles (IoEV).**

I. Introduction

THE INTERNET of Electric Vehicles (IoEV) aims to enhance the efficiency of Electric Vehicles (EVs) by enabling them to trade energy with each other as well as with Charging Stations (CSs). It has marked a significant shift in the transportation landscape, offering a more sustainable alternative to traditional fossil fuel-based vehicles [1]. However, the IoEV faces several challenges for energy trading between EVs and CSs that can hinder its full potential. These challenges include a lack of transparency, security concerns, inefficiencies due to intermediaries, limited peer-to-peer energy transactions, inadequate incentivization, and scalability issues [2]. To address these challenges, researchers and academicians are exploring the use of blockchain technology, with its inherent characteristics of decentralization, transparency, security, immutability, and smart contract functionality [3]. However, blockchain applications for energy trading are still developing and face several issues that need to be addressed [4]. The primary problem is the lack of a suitable consensus mechanism that supports the unique requirements of minimizing CPU workload, bandwidth consumption, energy inefficiencies, and excessive pending transactions [5]. Traditional consensus algorithms, including Proof-of-Stake (PoS) [6], [7], Proof-of-Authority (PoA) [8], [9], Practical Byzantine Fault Tolerance (PBFT) [10], [11], and Proof-of-Reputation (PoR) [12], while effective in their respective domains, do not adequately address the unique requirements for the energy trading in the IoEV. The constraints of these consensus methods include excessive energy use, high confirmation delays, low throughput capacity, poor scalability, and queued or long-pending transactions for processing and inclusion in the confirmed block. To address these issues, we propose RBCET, a reputation-based blockchain consensus mechanism for fast and secure energy trading in the IoEV. This mechanism is functional based on the Reputation Score (RS) of every EV and CS in the blockchain network. The EVs and CSs with RS values equal to or exceeding the average RS collectively are responsible for verifying and appending blocks to the blockchain. Among these EVs and CSs, the one with the highest RS is considered a leader. The proposed consensus mechanism divides time into epochs, where within each epoch, only one leader is accountable for generating blocks for the new set of transactions. Additionally, our proposed scheme includes the implementation of hardware wallets on each EV and CS to serve as a secure backup for the incentive mechanism to prevent unauthorized access effectively. The mathematical notations with their descriptions used in the proposed scheme are shown in Table I.

Received 11 August 2024; revised 28 December 2024; accepted 7 February 2025. Date of publication 14 February 2025; date of current version 14 August 2025. *(Corresponding author: Debasis Das.)*

Amritesh Kumar is with the Department of IDRP: IoT and Applications, Indian Institute of Technology Jodhpur, Jodhpur 342027, India (e-mail: amritesh.1@iitj.ac.in).

Koustav Kumar Mondal is with the Inter-Disciplinary Research Division: Centre for AIoT & Applications, Indian Institute of Technology Jodhpur, Jodhpur 342027, India (e-mail: mondal.4@iitj.ac.in).

Lokendra Vishwakarma is with the Department of Computer Science and Engineering, Thapar Institute of Engineering and Technology, Patiala 147004, India (e-mail: lokendra.vishwakarma@iitj.ac.in).

Debasis Das is with the Department of Computer Science and Engineering, Indian Institute of Technology Jodhpur, Jodhpur 342027, India (e-mail: debasis@iitj.ac.in).

Digital Object Identifier 10.1109/TCE.2025.3542172

1558-4127 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

TABLE I
KEY NOTATIONS

Notations	Descriptions	Notations	Descriptions
CU	CPU usage	MU	Memory usage
SoC	State-of-the-Charge	SoH	State-of-the-Health
NT	Network Throughput	IG	Information Gain
ET	Entropy	θ	Threshold
DTR	Decision Tree Regressor	MAE	Mean Absolute Error
MSE	Mean Squared Error		

A. Motivation

The Honda and General Motors led group has developed a global blockchain standard for EV and CS integration that lays the foundation for a worldwide decentralized vehicle charging network [13]. It highlights the potential of blockchain technology in decentralized energy management. However, the project faces significant challenges, such as security vulnerabilities, messaging standardization, and transaction automation in the IoEV framework. The state-of-the-art consensus mechanisms are used to solve these challenges but are energy-intensive and lack flexibility for real-time applications. Additionally, the absence of an effective incentive system in this project prevents EVs and CSs from engaging in energy exchanges. These challenges motivated us to develop a novel consensus algorithm named RBCET for fast and secure energy trading in the IoEV framework. Our approach also provides energy wallets into each EV and CS that serve as a secure backup for the incentive system and effectively prevent unauthorized access.

B. Research Contribution

The significant contributions of the proposed scheme in the field of blockchain-based IoEV are compiled as follows:

- We have proposed a novel reputation-based consensus mechanism called RBCET to address challenges such as intensive CPU workload, bandwidth consumption, energy inefficiencies, and excessive pending transactions associated with state-of-the-art approaches.
- The reputation score (RS) of each node is computed using its reputation parameters, such as CU, MU, NT (Mbps), SoH, and SoC.
- The proposed scheme utilizes a novel incentive computation method employing the RS and EnergyGiven parameters to enhance the efficiency of energy trading. It also uses an energy wallet to store and transfer the incentives to other nodes securely.
- We have used the decision tree regressor model to generate the reputation parameter and energy parameter datasets.
- The security analysis shows that the proposed scheme is secure from double spending, Distributed Denial of Service (DDoS), eclipse, Sybil, and collusion attacks. According to the outcomes of the performance analysis, RBCET reduces CPU workload by 47%, bandwidth utilization by 5%, energy consumption by 10%, pending transactions per block by 75%, and improves overall efficiency by 62%.

The remainder of the paper is structured as follows: Section II presents the literature survey. Section III describes network/adversary models and the design goals of the proposed approach. Section IV discusses the proposed, and Sections V and VI deal with security and performance analysis, respectively. Section VII is our conclusion and future work.

II. LITERATURE SURVEY

In this section, we review the most relevant state-of-the-art approaches that have explored the consensus mechanisms for energy trading in blockchain-enabled IoEVs. Sun et al. [4] proposed PBFT-based Delegated Proof-of-Stake (DPoSP) for increased energy sharing between EV-to-EV (E2E), EV-to-CS (E2C), and CS-to-CS (C2C) in IoEV. The challenges associated with this approach include heavy resource consumption in energy trading while maintaining a high consensus efficiency. To minimize resource consumption and improve security and performance in the context of electric vehicles to grid (E2G) power exchange systems, the authors of [14] proposed an enhanced Proof-of-Benefit (EPoB) consensus technique. The EPoB optimizes energy distribution that enables real-time adjustments in EV and CS charging and discharging schedules to flatten overall power load fluctuations. It faces significant challenges in real-time adaptive decision-making and scalable integration within existing grid infrastructures to handle the charging and discharging of EVs and CSs efficiently. The reputation-based delegated BFT enables smart communities to integrate distributed renewable energy sources with EVs while maintaining security and privacy [15]. However, the dynamic optimal contract assignment and energy allocation algorithm in energy production lead to challenges in the energy demands of EVs that affect the reliability and efficiency of the charging scheme [15]. Wang et al. [16] propose a consortium blockchain structure for energy trading along with a Block Alliance Consensus (BAC) algorithm that ensures data quality, dependability, and network security. The BAC faces challenges with scalability and robustness along with high communication overhead and computational demands necessary for real-time services in energy trading.

Abishu et al. [17] proposed E2E energy sharing within the IoEV using the PPoR consensus mechanism and a two-stage Stackelberg game-based approach featuring multiple leaders and followers. This approach aims to improve energy sellers' and purchasers' utility and income. The PPoR introduces validator selection dependency, cluster maintenance challenges, and higher communication complexity due to multiple rounds of interaction and reputation calculations among nodes for verifying the blocks within the clusters. Barnawi et al. [18] introduce a blockchain-based energy trading system that harnesses the potential of EVs and CSs in a green V2G environment. To evaluate the performance of this system, the authors employed two types of datasets: smart grid lab data [19] and US open energy information [20]. The movement of EVs and the varying availability of CSs create a highly dynamic environment. Consistent energy trading in

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON WITH STATE-OF-THE-ART CONSENSUS MECHANISMS FOR ENERGY TRADING WITH RBCET

Characteristics	[10], [11]	[6], [7]	[8], [9]	Proposed
Node Management	Private	Public	Consortium	Private
Transaction Latency	High	Low	Medium	Low
Throughput	Low	High	Medium	High
Energy Consumption	Low	Low	Low	Low
Fault Tolerence	1/3	1/2	1/2	1/3
Scalability	Poor	Rich	Rich	Rich

such fluctuating conditions is complex. Using game theory adds an extra layer of computational complexity that leads to inefficiencies when applied in real-time scenarios with high-frequency transactions. Some state-of-the-art approaches have also explored the use of blockchain technology in power trading systems among IoVs. For example, Javed et al. [21] developed an energy trading algorithm to manage energy demands and reduce superfluous requests. In this approach, the selling price of energy is determined by a smart payment contract. However, the presence of a large number of EVs in these systems brought about problems such as data redundancy and message credibility.

The intelligent consensus mechanism facilitates secure and efficient energy transactions between nodes, contributing to an optimized and sustainable ecosystem in smart cities. However, state-of-the-art consensus mechanisms encounter higher energy consumption, limited scalability, heavy CPU workload, and restricted throughput. These issues make them unsuitable for efficient and secure energy sharing between nodes. To overcome these issues and enable effective energy trading between nodes, we propose a novel consensus mechanism called RBCET. The proposed scheme is explicitly designed to address these issues in energy trading between nodes for smart city environments. Table II demonstrates the differences between the state-of-the-art consensus mechanisms frequently used in energy trading with RBCET.

III. PREREQIUSITE

In this section, we begin by introducing the network model under consideration, followed by the adversary model. Subsequently, we outline the design goals of our proposed schemes.

A. Network Model

The network model of the proposed scheme comprises three main components: EV, CS, and the Trusted Authority (TA). EVs are equipped with On-Board Units (OBUs) that enable communication with other EVs, CSs, and TA. From here on, we refer to EV and CS as blockchain nodes. Both of these nodes can be the buyer or seller. It is essential for nodes to have compatible hardware to facilitate energy trading with other nodes. The TA plays a crucial role by possessing computational capabilities to handle various IoEV tasks such as node registration, certificate issuance to nodes, and communication of node certificates to the blockchain. The network model's detailed description, depicted in Fig. 1, is outlined below:

1) All nodes must register with the Trusted Authority (TA) before participating in energy trading. The TA verifies the unique identity of nodes provided by the government before registration. If the unique identity is successfully verified, then TA registers the nodes. Once registered, each node receives a certificate from the TA, which they subsequently utilize to authenticate their legitimacy before participating in energy trading activities.
2) TA communicates the node's certificate on the blockchain network.
3) To initiate the energy trading process, every node communicates crucial information such as certificates, desired energy quantity, and energy cost to the blockchain. The blockchain then verifies the nodes' authenticity through certificates and examines the price difference between the expected energy cost and the current market price. If the node's authentication passes these criteria successfully, the blockchain permits the node to proceed with the energy trading; otherwise, it is denied.
4) When nodes intend to purchase energy, they relay their certificates, maximum energy requirement, and the price they are willing to pay to the blockchain. The blockchain then validates the nodes and shares the seller node's information with the buyer node, facilitating the energy purchase transaction.
5) Seller and buyer nodes engage in energy exchange, where the buyer node compensates the seller node with the agreed-upon incentive for the provided energy.
6) Seller node communicates the sold energy information to the blockchain.

B. Adversary Model

The adversary model describes the capabilities and limitations of potential adversaries who may attempt to manipulate the reputation score (RS) and thereby affect the consensus mechanism. We formulate the adversary model mathematically to capture these aspects.

1) Adversary Capabilities: Let $\mathcal{N}$ be the set of all nodes in the network, and $\mathcal{N}' \subseteq \mathcal{N}$ be the set of adversarial nodes. Each node $n \in \mathcal{N}$ has $RS(n)$, and each adversarial node $n' \in \mathcal{N}'$ has $RS(n')$.

- *Computational Power:* The adversary controls a fraction f of the total computational power in the network. The mathematical representation of computational power is shown in Equation (1).

$$f = \frac{\sum_{n' \in \mathcal{N}'} \mathrm{CU}(n')}{\sum_{n \in \mathcal{N}} \mathrm{CU}(n)} \tag{1}$$

- *Memory Manipulation:* The adversary can manipulate its memory usage to some extent δ to influence its reputation score. This is modeled in Equation (2).

$$\mathrm{MU}(n') = \mathrm{MU}(n') + \delta \tag{2}$$

- *Sybil Attacks:* The adversary can create s Sybil nodes to influence the consensus mechanism. The effectiveness of

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

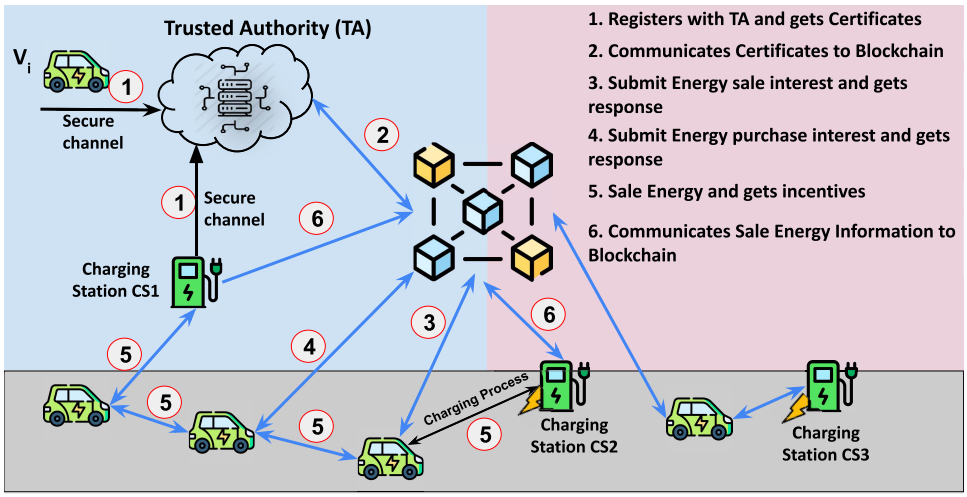


Fig. 1. Basic architecture of the proposed scheme.

this attack is limited by the reputation score shown in Equation (3).

$$s = \frac{RS(n')}{\theta} \tag{3}$$

2) Adversary Limitations: The adversary limitations are described using threshold limit, detection probability, and resource constraints parameters.

Threshold Limit: Adversarial nodes cannot participate in the consensus mechanism if $RS(n') < \theta$.

Detection Probability: The probability P_d of an adversarial node being detected and removed from the network is directly proportional to the extent of deviation Δ from the normal behavior. It is represented as $P_d = k \times \Delta, \quad 0 < k \leq 1$ where $\Delta = |RS(n') - RS_{\text{expected}}(n')|$.

Resource Constraint: The Equation (4) shows the adversary has limited resources to control the maximum number of nodes.

$$\textit{Resource Constraint}(n') = \frac{\text{Total_Resources}(n')}{\text{Resource_Per_Node}} \tag{4}$$

3) Adversary Objective: The primary objective of the adversary is to maximize its influence I on the consensus mechanism without being detected. This is mathematically formulated as $\max_{n' \in \mathcal{N}'} I(n') = \alpha \times RS(n') - \beta \times P_d$. Here, $RS(n') \geq \theta$ and $P_d \leq P_{\max}$, α and β are weighting factors, and $P_{\max}$ is the maximum tolerable detection probability.

C. Design Goal

The primary design goal of the proposed scheme is to establish a robust and reliable consensus mechanism among blockchain nodes. The proposed scheme aims to achieve the following specific objectives.

Quantitative Trustworthiness and Dynamic Participation: The proposed scheme facilitates a reliable consensus process to support a quantitative measure of a node's trustworthiness through their RS value. It dynamically decides the node eligibility to participate in the consensus mechanism based on the RS value. The RS value of the node should be greater than the θ to satisfy the eligibility. The higher RS value improves the security and reliability of the blockchain network by allowing only reputed nodes to participate in the consensus and transaction processes.

Realistic Dataset Generation: The dataset is generated in the proposed scheme by establishing mathematical relationships among various node parameters such as CU, MU, SoH, SoC, and NT.

Hardware-Based Incentives: The hardware-based incentive mechanism in the proposed scheme rewards nodes based on their contributions to the network which encourages better performance and participation.

Scalability and Efficiency: The proposed scheme uses machine learning models, specifically the DTR, for efficient RS calculation that ensures the system scales well with an increasing number of nodes.

By achieving these design goals, the system aims to create a more secure, efficient, and trustworthy blockchain network.

IV. Proposed Approach: RBCET

The proposed scheme calculates the RS value of each node to decide their participation in the consensus mechanism. The nodes whose RS value is greater than the θ are allowed to participate in the consensus process and trade the energy. The value of θ equals the average value of the RS of all nodes. Additionally, the hardware-based incentive mechanism is used to calculate the incentive for the blockchain nodes.

A. Node's Reputation Parameters

In the proposed scheme, we define specific mathematical correlations between the reputation parameters of each node to create a more realistic dataset. Here's an explanation of how we form these relationships:

- CU and MU: We assume a direct correlation between CU and MU because when CU rises, MU will also increase. The Equation (5) represents the relationship between CU

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

TABLE III
REPUTATION PARAMETER DATASET

SNo.	CU	MU	NT	SoH	SoC
0	2	4	17	66	27
1	85	115	60	67	22
2	64	84	49	68	8
...	...	...	...	...	...
199997	69	92	52	66	35
199998	7	2	19	66	36
199999	61	80	47	67	19

and MU.

$$MU = \sum_{i=0.5}^{1.5} [k_i * CU] + \sum_{j=-10}^{10} b_j \tag{5}$$

Here, k and b are randomly generated variables whose values fall within a certain range to create variation in the data.

- SoH and SoC: We assume that there exists a correlation between the SoH and SoC such that as the SoC decreases, the SoH also decreases. The relationship between SoH and SoC using a quadratic equation is shown in Equation (6).

$$soh = \sum_{i=0.001}^{0.005} \left[a_i * soc^2\right] + \sum_{j=-0.5}^{-0.1} \left[b_j * soc\right] + \sum_{50}^{80} c_k \tag{6}$$

Here, the values of a,b, and c are randomly generated within a certain range to determine the shape of the curve.

- NT and CU: We assume a positive correlation between CU and NT. Here, the NT increases to a certain threshold when the network becomes saturated. Equation (7) and (8) represents this relationship.

$$X = e^{\sum_{i=0.01,j=40}^{0.05,60}(-p_i*(cpu_usage-q_j))} \tag{7}$$

$$NT = \sum_{l=80}^{120} \frac{k_l}{1+X} \tag{8}$$

The values of k, p, and q are randomly generated within a certain range.

B. Reputation Score (RS) Calculation

The proposed scheme calculates the RS value of each node using its parameters to ensure that the network remains reliable, secure, and trustworthy. We constructed the dataset based on the relationships and values of the node's parameters. Table III shows the reputation parameter dataset based on CU, MU, NT (Mbps), SoH, and SoC parameters of nodes.

The RS provides a quantitative measure of a node's trustworthiness that makes it easier for the blockchain network to reach a consensus and function smoothly. As shown in Table IV, we have taken eight different machine learning models in benchmarking. We have selected the DTR model because the value of MAE & MSE is less than other machine learning models. The DTR model also offers several key advantages for the proposed application. ① It provides a human-interpretable structure that enables domain experts to visualize how different parameters (CU, MU, NT, SoH, SoC) interact to influence the RS value. Unlike complex black-box models, the DTR explicitly encodes the logic used in each prediction step. ② DTR handles nonlinear relationships and complex parameter interactions effectively which allows it to model subtle nuances in how resource consumption or network throughput affects trustworthiness. ③ DTR supports feature selection via IG that ensures that only the most predictive parameters drive the model splits which increases efficiency and interpretability. Figure 2 depicts the Decision Tree structure as induced from our parameter dataset. The tree selects the feature that provides the highest IG for RS at the top level. The Equation (9) computes the IG that uses an ET-based measure where ET quantifies uncertainty or impurity in the RS distribution. The ET is computed using the Equation (10).

$$IG = ET_P - [AVG_{wt}(ET_C)] \tag{9}$$

$$ET = -\sum_i p_i \log_2(p_i) \tag{10}$$

Here, p_i represents the proportion of data points (nodes) falling into a particular outcome category. To determine the best splitting feature, we compute the IG based on parent and child ET. The ET_P is the entropy of the parent node, and ET_C denotes the entropy of each child node after a potential split. The feature that results in the highest IG reduces the uncertainty in RS and is thus chosen for the split.

Figure 2 shows that the DTR model starts from the root and might split on NT, then proceed to MU or CU, and so forth, selecting the most informative feature at each step. It also shows the example regression lines at intermediate nodes that illustrate how the chosen splitting feature relates to the RS value. For instance, if splitting on NT yields the highest information gain (IG), the tree first divides the dataset into subsets where NT is below or above a certain threshold. The subsequent splits involve MU or CU if those features offer the highest additional IG in the resulting subsets. As the tree grows deeper, it refines its partitions by focusing on parameter combinations that produce consistent and predictable RS values. Each tree branch leads to a further division based on another feature until reaching a leaf node. A leaf node provides a final predicted RS value for the subset of data it represents. The process continues till the stopping criteria, such as maximum tree depth, a minimum number of samples per leaf node, or a sufficiently low impurity level, are met. The final leaves, annotated with `Reputation Score = ...`, describe the predictive power of the tree since it isolates combinations of parameters that yield stable and accurate RS estimates. Once trained, the resulting model is applied to predict the RS value for any new node by traversing the tree from the root to a leaf and following the parameter-based decision thresholds at each node. Although the DTR model partitions the input space piecewise, one can also model the RS with a parametric equation learned from the data. While Equation (11) provides a parametric form that emerged from fitting a model to the dataset, the DTR model encapsulates this relationship through hierarchical splits rather than relying on a single closed-form equation. The constants $x_1, x_2, x_3, \ldots, x_8$ in this equation represent learned parameters

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
BENCHMARKING

Model Name	MAE	MSE	RMSE	R2	RMSLE	MAPE	ET(Sec)
K Neighbors Regressor	0.0057	3.400000e-03	0.0579	1.0000	0.0007	0.0001	1.102
Extra Trees Regressor	0.0045	2.300000e-03	0.0480	1.0000	0.0007	0.0001	32.703
Decision Tree Regressor	**0.0044**	**2.300000e-03**	**0.0480**	**1.0000**	**0.0007**	**0.0001**	**0.338**
CatBoost Regressor	0.1908	5.820000e-02	0.2412	1.0000	0.0019	0.0014	24.867
Light Gradient Boosting Machine	0.2027	6.920000e-02	0.2631	1.0000	0.0021	0.0014	1.333
Random Forest Regressor	0.0045	2.300000e-03	0.0480	1.0000	0.0007	0.0001	24.764
Gradient Boosting Regressor	0.3012	1.388000e-01	0.3725	1.0000	0.0030	0.0021	11.096
Extreme Gradient Boosting	0.1611	4.970000e-02	0.2228	1.0000	0.0018	0.0011	17.156

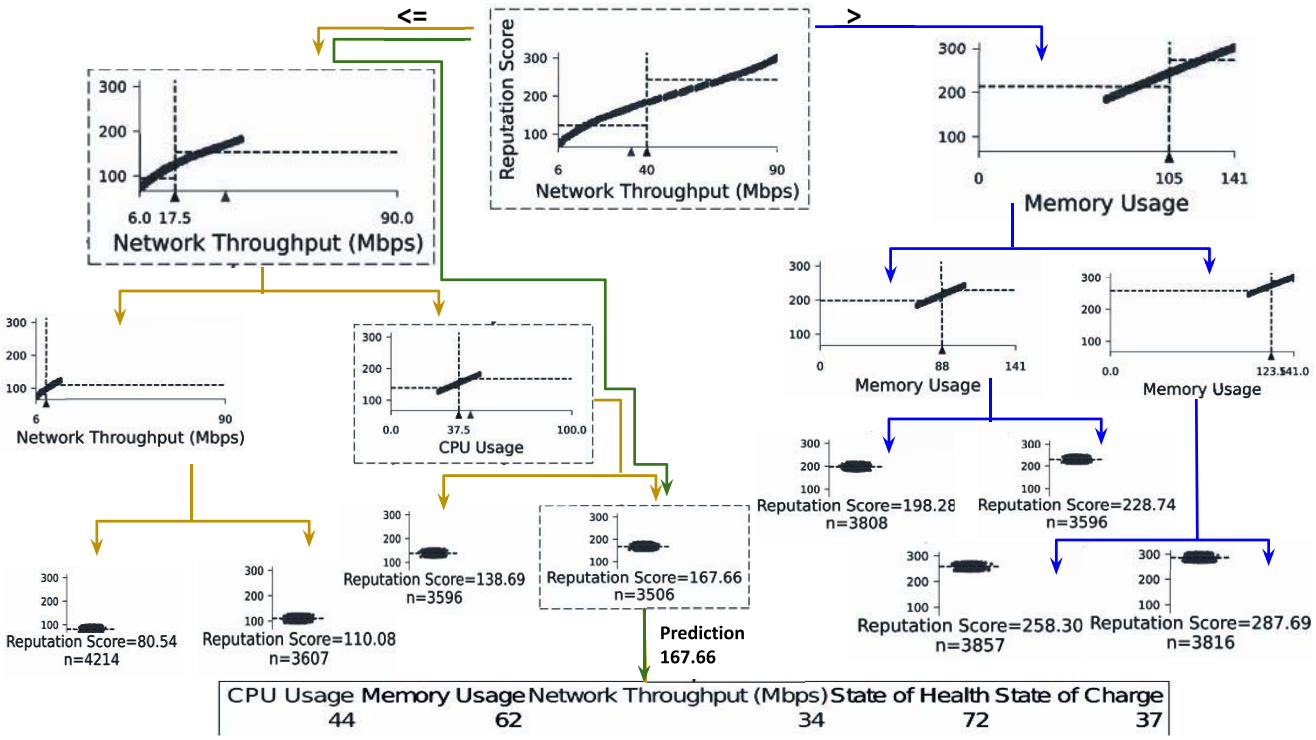


Fig. 2. Decision tree structure for RS prediction. Each node represents a condition on a particular feature, and each edge represents a branch based on that condition. The tree refines the subset of parameters until it reaches leaf nodes that give predicted RS values.

that reflect how each input, including MU, CU, NT, State-of-the-Charge (SoC), and State-of-the-Health (SoH), affects the RS value. These interactions are approximated through a structured approach that leverages feature selection and hierarchical partitioning by training the DTR model. The DTR delivers accuracy and interpretability that make it a strong choice for quantifying node reputation in blockchain environments. The DTR model systematically evaluates parameters, selects the most informative features, and constructs a hierarchical model that is easily visualized and understood. This allows network operators and stakeholders to trust the metric and the process by which the RS value of each node RS is derived.

$$RS = \frac{x_1 \cdot MU + x_2 + x_3 \cdot (SoC)^2 + x_4 \cdot SoC + x_5 + x_6}{1 + \exp(-x_7 \cdot (CU - x_8))} \quad (11)$$

1) Example of RS Calculation: This section shows a step-by-step calculation of the RS value using the first row from the reputation parameter dataset as shown in Table III. The constants used in the equation are obtained from a previously trained model. The first row contains the value of the reputation parameter such as CU = 2, MU = 4, NT = 17, SoH = 66, and SoC = 27. The value of the constants $(x_1, x_2, \ldots, x_8)$ get from training is $x_1 = 0.5, x_2 = 10, x_3 = 0.01, x_4 = -0.05, x_5 = 3, x_6 = -2, x_7 = 0.1$, and $x_8 = 50$. The $(SoC)^2$ is computed as $(SoC)^2 = (27)^2 = 729$. Substitute into the numerator of Equation (11) as $x_1 \cdot MU = 0.5 \times 4 = 2$, $x_3 \cdot (SoC)^2 = 0.01 \times 729 = 7.29$, $x_4 \cdot SoC = -0.05 \times 27 = -1.35$. Combine all terms in the numerator as $2 + 10 + 7.29 - 1.35 + 3 - 2 = 18.94$. The denominator is $1 + \exp(-x_7(CU - x_8))$. Substitute $CU = 2$, $x_7 = 0.1$, and $x_8 = 50$ in denominator and we get ≈ 122.51. Now we have $RS = \frac{18.94}{122.51}$ which is ≈ 0.1547.

C. RBCET Consensus Algorithm

RBCET improves consensus efficiency by using the RS value calculations. RBCET uses a node to validate the blocks whose RS is greater than or equal to θ, so non-reputed nodes do not participate in validating the blocks. The node with the highest RS value in one epoch is called the leader node. The pseudocode for RBCET is shown in Algorithm 1.

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Consensus Algorithm Based on RS

Input: N:List of nodes, RS:Reputation Score of each node, T:Timestamp, E_{inf}:Energy Information
Output: Selected set of nodes S for consensus

1: Calculate θ as the median of RS for all nodes in N
2: Initialize $S = \emptyset$
3: **for** each node n in N **do**
4: **if** $RS(n) > \theta$ **then**
5: $S = S \cup \{n\}$
6: **end if**
7: **end for**
8: LN $\leftarrow RS_{max}(S)$ and RN $\leftarrow S \setminus LN$
9: **if** (pre-prepare, $seq_{no.}$, $H(E_{inf})$, T) == true **then**
10: communicate (pre-prepare,$seq_{no.}$,$H(E_{inf})$,T)$sign_{LN}$)
11: receive (pre-prepare, $seq_{no.}$,$H(E_{inf})$, T)$sign_{RN}$))
12: **else**
13: "Do-Nothing"
14: **end if**
15: **if** (prepare, $seq_{no.}$, $H(E_{inf})$, S, T) == True **then**
16: communicate (prepare, $seq_{no.}$, $H(E_{inf})$, S, T) $sign_S$)
17: receive (prepare, $seq_{no.}$, $H(E_{inf})$, S, T) $sign_S$)
18: **else**
19: "Do-Nothing"
20: **end if**
21: **if** (response, $H(E_{inf})$, $seq_{no.}$, S, T) == True **then**
22: count = count+1
23: **if** count $\geq$ 2f+1 **then**
24: broadcast (reply, $H(E_{inf})$, $seq_{no.}$, T)
25: **else**
26: "Do-Nothing"
27: **end if**
28: **end if**

The algorithm aims to achieve consensus in a decentralized network by selecting a subset of nodes based on their RS value. It introduces a two-tier hierarchy among the chosen nodes that designates one as the Leader Node (LN) and the rest as Reputed Nodes (RN). This hierarchy is used to facilitate and validate the consensus process. Line 1 calculates the average of the RS (θ) of all nodes in N. An empty set S is initialized to store the selected nodes in line 2. Lines 3–7 choose the nodes based on the RS value. If the RS value of node n $>$ θ then it is added to the set S. The node with the maximum RS value in set S is designated as the LN whereas others are designated RN in line 8. The pre-prepare phase is described in lines 9–14. The LN initiates the pre-prepare phase if certain conditions are met. A signed message is communicated to all RNs. Upon receiving the pre-prepare message, the RNs enter the prepare phase if conditions are met from lines 15–20. A signed message is communicated and received among the nodes in S. Nodes in S count the number of received prepared messages. If the count exceeds a threshold of 2f+1, a reply is broadcast to finalize the consensus. By following these steps, the algorithm ensures that only nodes with a high reputation score participate in the consensus process.

D. Hardware-Based Incentive Mechanism

The hardware-based incentive mechanism in the proposed approach is not just a feature but a necessity. It ensures active participation, encourages honest behavior, and makes the network economically sustainable. The proposed scheme determines the incentive through Equation (12), considering both the RS and the energy a node provides to another node. The incentive is proportional to the RS and the energy given, with nodes with a higher RS value offering more energy receiving a greater incentive.

$$Incentive = \frac{RS}{RS_{max}} * \frac{EnergyGiven}{MaximumEnergy} \tag{12}$$

The maximum RS value (RS_{max}) is the maximum possible value of RS that any node can achieve. The energy the node gives to another node is defined as the energy given, measured in units such as kWh. The maximum energy is the maximum possible energy that any node can provide to other nodes. The system predetermines the RS_{max} and maximum energy to establish the maximum level of accuracy and reliability required for participation and the maximum amount of energy that can be traded. The well-defined incentive formula helps ensure the system's security and fairness while enabling easy involvement and integration with existing infrastructure. The incentive is securely stored and transferred to other nodes using a hardware wallet, such as an energy wallet. Each node is equipped with an edge device to implement the energy wallet. The edge device is connected to sensors and controllers that monitor the node's battery state and control the energy flow from one node to another. Therefore, incorporating a hardware-based incentive mechanism is crucial for the long-term success and reliability of the blockchain network.

V. Security Analysis

This section explains the security proof of the proposed scheme. As mentioned in the previous sections, the scheme is protected from double-spending, liveness, eclipse, Sybil, and collusion attacks. Below, we have described in detail how the RBCET is protected from all these attacks. All honest nodes have the same view after expansion. We utilize Python as the primary programming language for implementing the blockchain and attack simulations due to its extensibility and rich ecosystem of libraries. The core dependencies include `numpy` for high-performance mathematical operations and custom-built modules such as `blockchain.py` for blockchain structure and transaction handling. The `monitor.py` is used for real-time blockchain monitoring, and `simulate_attacks.py` is used for attack orchestration driver. The attack-specific modules like `ddos.py` for DDoS attack simulation,`double_spending.py` for double-spending prevention test, `eclipse.py` for peer isolation simulation, `sybil.py` for fake identity injection, and `collusion.py` for multi-vehicle collusion detection. The setup leverages `matplotlib` for visualizing blockchain states. The attack affects and operates on Linux-based environments for optimized networking, concurrency, and scalability in testing distributed attack scenarios.

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

Lemma 1: Let N be the set of all nodes in the blockchain network, and $RS(i)$ be the Reputation Score of node i. If every node i participating in the consensus process satisfies $RS(i) > \theta$, where θ is the average Reputation Score threshold, then the probability $P(DS)$ of a successful double-spending attack is negligible, $P(DS) < \epsilon$, where ϵ is a small constant approaching zero.

Proof: Let p denote the probability of a randomly selected node satisfying $RS(i) > \theta$. We assume an adversary controls a number $N_a \in N$ of nodes in the network. Each adversary node $i \in N_a$ attempts to double-spend by validating conflicting transactions. The adversary nodes must ensure that most of the nodes in the consensus process are malicious to do the adversary process. Let $|N_c|$ be the number of nodes required to reach consensus and N_h represent the number of honest nodes. So, $N = N_a + N_h$. The probability P_s of the adversary controlling the consensus process is $P_s = \binom{|N_c|}{|N_a|} \cdot (p_a)^{|N_a|} \cdot (1-p_a)^{|N_c|-|N_a|}$ where p_a is the probability of an adversarial node satisfy the $RS > \theta$. The $\theta = \frac{1}{|N|}\sum_{i \in N} RS(i)$ and $|N|$ is the total number of nodes. The RS of a node i follows two distinct distributions. First, $RS_h \sim F_h(x)$ for honest nodes that tend to be higher on average, and second, $RS_a \sim F_a(x)$ for adversarial nodes that tend to be lower on average. The cumulative distribution function (CDF) of adversarial scores is $P(RS_a \leq x) = F_a(x)$, and $P(RS_a > x) = 1 - F_a(x)$. The RS value must exceed θ of the adversarial nodes to participate in consensus. The $P(RS_a > \theta) = 1 - F_a(\theta)$. The adversarial probability $P(RS_a > \theta)$ decreases by substituting θ because θ is dominated by higher RS_h values of honest nodes. The expected number of adversarial nodes passing θ for N_a adversarial nodes is $\mathbb{E}[N_{a,c}] = N_a \cdot (1 - F_a(\theta))$. Since $F_a(\theta)$ is high due to low RS_a scores relative to honest scores, $(1 - F_a(\theta))$ becomes negligible, minimizing $\mathbb{E}[N_{a,c}]$.The N_c participates in the consensus with $N_c = N_h + N_{a,c}$. The fraction of adversarial nodes in consensus is $f_a = \frac{N_{a,c}}{N_c} = \frac{\mathbb{E}[N_{a,c}]}{N \cdot (1 - F_a(\theta))}$. As $F_a(\theta) \to 1$, $(1 - F_a(\theta)) \to 0$, making f_a approach zero such as $\lim_{F_a(\theta) \to 1} f_a = 0$. Since $f_a \to 0$, the consensus process remains dominated by honest nodes N_h, ensuring that adversarial influence is negligible. For an adversarial majority to succeed, $N_{a,c}$ must exceed $\frac{N_c}{2}$. The probability $P_{Adversarial\ Majority} = P(N_{a,c} > \frac{N_c}{2})$. Given $N_{a,c}$ follows a binomial distribution with N_a trials and success probability $1 - F_a(\theta)$, we write $P(N_{a,c} = k) = \binom{n_a}{k} \cdot (1 - F_a(\theta))^k \cdot (F_a(\theta))^{n_a - k}$. For $N_{a,c} > \frac{N_c}{2}$, the cumulative probability is $P(N_{a,c} > \frac{N_c}{2}) = \sum_{k=\lceil N_c/2 \rceil}^{N_a} \binom{N_a}{k} \cdot (1 - F_a(\theta))^k \cdot (F_a(\theta))^{N_a - k}$. The $P(N_{a,c} > \frac{N_c}{2}) \to 0$ because $(1 - F_a(\theta)) \to 0$. The $P_{validation}$ represents the probability that a malicious transaction passes validation $P_{validation} = (1 - q)_h^N$ where q is the detection probability. The total probability of a successful double-spending attack is $P(DS) = P_{Adversarial\ Majority} \cdot P_{validation}$. Since both terms are negligible, the $P(DS)$ is $P(DS) \to 0$. The simulation result in Fig. 3(b) shows that the RBCET is secured from Double Spending attack. ∎

Lemma 2: If all nodes i in the network satisfy $RS_i > \theta$ and $RL_i > \alpha$, then the probability of a successful DDoS attack $P(\text{DDoS})$ is less than a negligible ϵ.

Proof: Let's assume we have a network with N nodes, and each node i has a Reputation Score RS_i and a Rate Limit RL_i. The system uses these two metrics to filter incoming traffic. Specifically, a node only processes an incoming request if $RS_i > \theta$ and $RL_i > \alpha$, where θ and α are predefined thresholds. Here, $P(\text{DDoS})$ is the probability of a DDoS attack being successful. $P(RS_i > \theta)$ is the probability that a node i has $RS_i > \theta$. $P(RL_i > \alpha)$ is the probability that a node i has $RL_i > \alpha$. To prove $P(\text{DDoS}) < \epsilon$, we calculate the individual probabilities $P(RS_i > \theta)$ and $P(RL_i > \alpha)$ for a single node i where $P(RS_i > \theta) = p_1$ and $P(RL_i > \alpha) = p_2$. Then, we calculate the joint probability for all nodes involved where $P(RS_i > \theta \text{ and } RL_i > \alpha, \forall i \in N) = p_1 \times p_2$. Given the conditions, the probability of a DDoS attack $P(\text{DDoS})$ is reduced where the condition is $P(\text{DDoS} | RS_i > \theta \text{ and } RL_i > \alpha, \forall i \in N) < P(\text{DDoS})$. Prove that $P(\text{DDoS} | RS_i > \theta \text{ and } RL_i > \alpha, \forall i \in N) < \epsilon$ we use $P(DDoS | RS_i > \theta RL_i > \alpha, \forall i \in N) = P(\text{DDoS}) \times (1 - p_1 \times p_2) < \epsilon$. Since $p_1 \times p_2$ can be made arbitrarily close to 1 by increasing p_1 or p_2, $P(\text{DDoS} | RS_i > \theta \text{ and } RL_i > \alpha, \forall i \in N)$ can be made smaller than ϵ. The lemma proves that if the proposed approach of using Rate Limiting and Reputation Scores is employed. The conditions $RS_i > \theta$ and $RL_i > \alpha$ are maintained for all i, then $P(\text{DDoS}) < \epsilon$, making the system secure against DDoS attacks. The simulation result in Fig. 3(c) shows that the RBCET is secured from a DDoS attack. ∎

Lemma 3: In a network N with H honest nodes and A adversarial nodes, where each node n maintains a peer list $P(n)$ of size k, if $A < \frac{k}{2}$, then the network is secure against eclipse attacks.

Proof: Here, N: Total number of nodes in the network. H: Number of honest nodes. A: Number of adversarial nodes. k: Size of the peer list for each node. $P(n)$: Peer list of node n. We assume that nodes choose peers uniformly randomly from N and the attacker cannot spoof identities. We demonstrate the proposed scheme's security against eclipse attacks by establishing that $A < \frac{k}{2}$. The random peer selection shows the probability p of any node n selecting an adversarial node a as a peer is $p = \frac{A}{N}$. The binomial distribution describes the number of adversarial nodes in $P(n)$ follows a binomial distribution $B(k, p)$. The eclipse condition demonstrates that for an eclipse attack to occur, the number of adversarial nodes in $P(n)$ must be greater than $\frac{k}{2}$ where $|P(n) \cap A| > \frac{k}{2}$. The probability of eclipse using the binomial distribution is the probability P_E of $|P(n) \cap A| > \frac{k}{2}$ can be calculated as $P_E = \sum_{i=\lceil \frac{k}{2} \rceil}^{k} \binom{k}{i} p^i (1-p)^{(k-i)}$. The threshold condition demonstrates that our lemma states if $A < \frac{k}{2}$, then P_E should be negligible. The negligible probability describes that given $A < \frac{k}{2}$, we have $p < \frac{1}{2}$. In this case, using the properties of the binomial distribution, it can be shown that P_E approaches zero as N grows large, making it negligible. Since P_E is negligible under the condition $A < \frac{k}{2}$, we conclude that the blockchain network is secure against eclipse attacks. The simulation result in Fig. 3 shows that the RBCET is secured from an Eclipse attack. ∎

Lemma 4: If all nodes participating in the consensus mechanism have an RS value greater than a dynamically computed

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

```
[Simulating Double-Spending Attack]
Legitimate transaction added:
vehicle1->vehicle2|Charge: 30
Double-spending attack prevented:
Double-spending attempt detected
for Vehicle1.
Double-Spending Attack Resisted.
=== Blockchain Monitor ===
[Blockchain Details]
Block 0:
Hash: fe1792572b9287c4alcf12377bfcc
4b0186d4c8a8702739344abe8dce24e74dc
Previous Hash: 0
Transactions (0):
```

(a) Double-Spending Attack

```
(Simulating DDoS Attack]
Simulating DDoS attack...
Transaction 1 failed: Vehicle1 cannot trade
with Vehicle2 as it has equal or lower charge.
Transaction 2 failed: Vehicle1 cannot trade
with Vehicle2 as it has equal or lower charge.
Transaction 3 failed: Vehicle1 cannot trade
with Vehicle2 as it has equal or lower charge.
Transaction 4 failed: Vehicle1 cannot trade
with Vehicle2 as it has equal or lower charge.
.....
.....
Transaction 49 failed: Rate limit exceeded.
Request dropped.
Transaction 50 failed: Rate limit exceeded.
Request dropped.
DDoS Attack Simulation Completed.
```

(b) DDoS Attack

```
Simulating eclipse attack...
Eclipse attack detected: Network
isolation or insufficient peers.
Eclipse attack prevented: Peer
validation failed.
Eclipse Attack Resisted.
=== Blockchain Monitor ===
[Blockchain Details]
Block 0:
Hash:fe1792572b9287c4alcf12377bfc
c4b0186d4c8a8702739344abe8dce24e7
4dc, Previous Hash:0
Transactions (0):
```

(c) Eclipse Attack

Fig. 3. Security attacks.

```
Simulating Sybil attack...
Node sybil1 rejected.
Node sybil2 rejected.
Node sybil3 rejected.
Invalid peer detected:sybil1
Sybil attack prevented: Peer
validation failed.
Sybil Attack Resisted.
=== Blockchain Monitor ===
[Blockchain Details]
Block 0:
Hash:fe1792572b9287c4alcf123
77bfcc4b0186d4c8a8702739344a
be8dce24e74dc,Previous Hash:0
Transactions (0):
```

(a) Sybil Attack

```
[Simulating Collusion Attack]
Simulating Collusion Attack...
No collusion detected for
Vehicle1->Vehicle4.
Malicious transaction:
Vehicle1->Vehicle4|Charge:2
No collusion detected for
Vehicle2->Vehicle4.
Malicious transaction:
Vehicle2 -> Vehicle4|Charge:3
Repeated malicious transaction:
Vehicle1->Vehicle4|Charge:7
Collusion attack prevented:
Collusion detected: 3 senders
targeting Vehicle4.
Collusion Attack Resisted.
```

(b) Collusion Attack

Fig. 4. Specific attacks to reputation-based system.

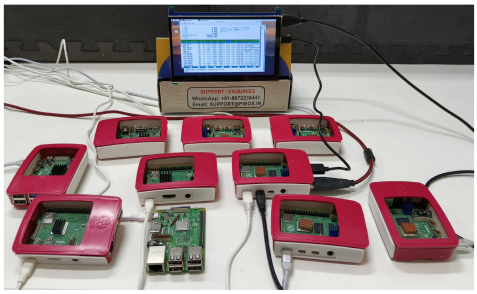


Fig. 5. Testbed setup.

threshold θ, the probability of a successful Sybil attack in the RBCET system is negligible.

Proof: Let N be the total number of nodes in the network. N_{honest} and N_{sybil} represent the number of honest and Sybil nodes where $N = N_{\text{honest}} + N_{\text{sybil}}$. $RS(n)$ denotes the Reputation Score of node n. θ is the threshold such that $RS(n) \geq \theta$ for a node n to participate in the consensus.p is the probability that a randomly chosen node has $RS(n) \geq \theta$. We assume that the RS of each node is independently and identically distributed. The probability that a node exceeds the threshold θ is p so $p = P(RS(n) \geq \theta)$. The Sybil attacker needs to have a significant fraction of nodes with $RS(n) \geq \theta$ to influence the consensus mechanism. If s is the number of Sybil nodes needed to impact the consensus significantly, then the probability of having s Sybil nodes each with $RS(n) \geq \theta$ is computed by the binomial distribution $P_{\text{sybil}}(s) = \binom{N_{\text{sybil}}}{s} p^s (1-p)^{N_{\text{sybil}}-s}$. The $\binom{N_{\text{sybil}}}{s}$ is the binomial coefficient that represents the number of ways to choose s successful nodes from N_{sybil}. p is the probability that a single node's RS exceeds θ. p^s is the probability of s nodes each independently exceeding θ. $(1-p)^{N_{\text{sybil}}-s}$ accounts for the remaining $N_{\text{sybil}} - s$ nodes not exceeding θ. The formula shows that as N_{sybil} increases, the number of combinations $\binom{N_{\text{sybil}}}{s}$ increases, but the probability of each combination $p^s(1-p)^{N_{\text{sybil}}-s}$ becomes increasingly unlikely if p is small. For the proposed scheme where θ is chosen based on the average of the total RS value, p is expected to be low for potential Sybil nodes not consistently contributing positively to the network. Hence, as N_{sybil} increases, $P_{\text{sybil}}(s)$ becomes negligibly small for any non-trivial s (i.e., $P_{\text{sybil}}(s) \to 0$ as $N_{\text{sybil}} \to \infty$). This analysis validates the lemma by demonstrating that the probability of a successful Sybil attack is negligible based on the threshold θ such that only reputed nodes participate in the consensus process. The simulation result in Fig. 4(b) shows that the RBCET is secured from Sybil attack. ■

Lemma 5: The RBCET reduces the probability of successful collusion through randomized selection, reputation score, and decentralized verification that ensures the collusion is both detectable and improbable.

Proof: Let γ be the factor adjusting the RS value of the nodes for selection probability and s be the size of a colluding group trying to influence the consensus. Each node's selection probability for a consensus mechanism is weighted by its RS value raised to a power γ and normalized across all nodes using $P_{\text{selected}}(n) = \frac{RS(n)^\gamma}{\sum_{i=1}^{N} RS(i)^\gamma}$ formula. We assume that the selections are independent and the probability that the s colluding nodes is selected in a given round is the product of their individual probabilities equivalent to $P_{\text{group}} = \prod_{j=1}^{s} P_{\text{selected}}(n_j)$. The colluders need to control the consensus across multiple rounds to attack in the network. The proposed scheme controls m out of M total rounds using the binomial distribution $P_{\text{success}} = \sum_{i=m}^{M} \binom{M}{i} P_{\text{group}}^i (1 - P_{\text{group}})^{M-i}$. Here, M is the total number of consensus rounds, and m is the minimum number of rounds the colluders need to control. The reputation adjustment factor γ is used to ensure that no small group of nodes can easily dominate the selection process unless they genuinely have the highest RS value. The probabilistic measure P_{success} becomes negligible as s

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

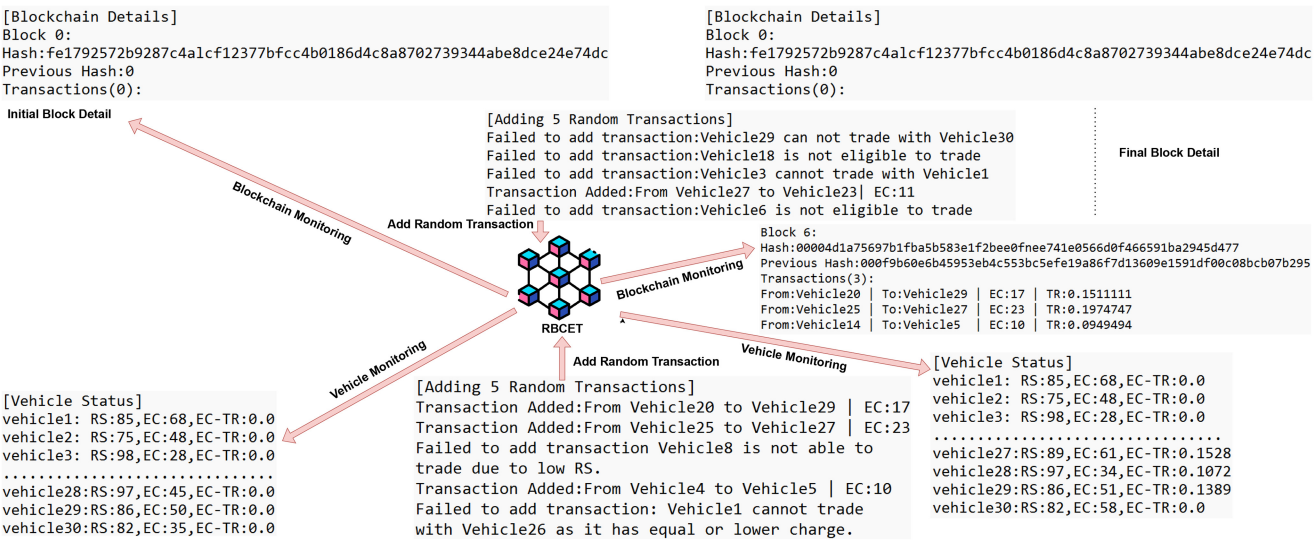


Fig. 6. Simulation flow.

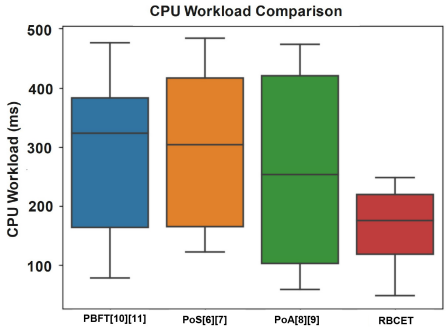


Fig. 7. CPU workload.

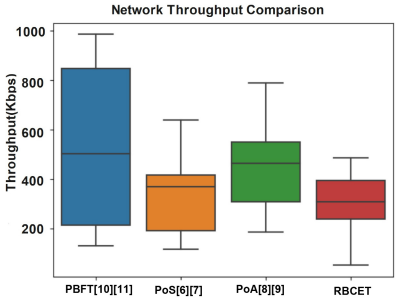


Fig. 8. Bandwidth utilization.

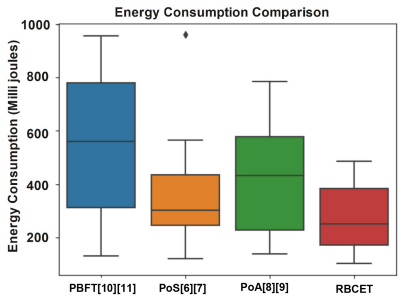


Fig. 9. Energy consumption.

increases or as γ increases making it computationally infeasible for colluders to maintain control over necessary consensus rounds without being detected or counteracted by other honest nodes. While colluders succeed in influencing a round, the decentralized verification by other nodes across the network serves as verification against falsified or malicious transactions being approved. This mathematical approach shows that by using γ and setting k, m, and M, the proposed scheme is secure from a collusion attack. The simulation result in Fig. 4 shows that the RBCET is secured from a collusion attack. ■

VI. Performance Analysis

The RBCET implementation details and performance assessment are described and summarized in this section. The effectiveness of implementation is fully proportional to the device configuration and algorithm. RBCET has been tested on devices such as Raspberry Pi 4 Model B. The device is categorized as an electric vehicle (EV) and a charging station (CS). The device platform structure includes a quad-core 64-bit Broadcom 2711 Cortex A72 Processor, WLAN 802.11b/g/n/ac (2,4 + 5,0 GHz), and LAN RJ45 10/100/1000 Mbit connectivity (Gigabit LAN over USB 3.0). Spyder 4.1.5 Python IDE on Anaconda Navigator and Flask 0.12.2 are the software setups. The blockchain network is constructed on CS using Python programming. We executed the whole procedure, which included the creation of four CS and six EVs. Each CS and EV connects to any other CS and EV for energy trading. The efficiency of the RBCET has improved compared to the state-of-the-art schemes in the context of CPU workload, bandwidth utilization, energy consumption, and pending transactions per block. Fig. 5 shows the testbed setup and benchmark dashboard, and Fig. 6 demonstrates the simulation flow of RBCET.

A. CPU Workload

The consensus mechanism is more easily scaled to accommodate a growing number of network participants by reducing the CPU workload. This allows the network to handle more

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.

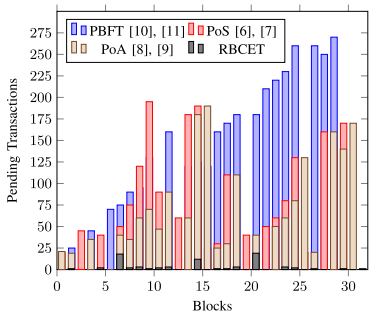


Fig. 10. Pending transactions per block.

transactions and support a larger number of users without sacrificing performance. The reduced CPU workload leads to improved performance for consensus mechanisms by improving network scalability. The RBCET reduces the CPU workload by using a consensus score to evaluate the nodes. As the simulation results are shown in Fig. 7, the RBCET scheme reduces the CPU workload more than PBFT [10], [11], PoS [6], [7], and PoA [8], [9] by 456.39%, 50.09%, and 12.75%, respectively.

B. Bandwidth Utilization

Lowering bandwidth utilization is an important factor in improving the performance of consensus mechanisms. It usage reduces the latency, or delay, in communication between nodes. This can make the consensus mechanism more responsive, improving the overall performance. It also reduces the costs associated with running a consensus mechanism. This can make it more accessible to a wider range of users and organizations. The RBCET uses lower bandwidth since the communicated message sizes are very small for energy trading. The simulation result in Fig. 8 shows that it reduces the bandwidth utilization than PBFT [10], [11], PoS [6], [7], and PoA [8], [9] by 54%, 10%, and 34%.

C. Energy Consumption

Lowering energy consumption can help improve the performance of the consensus mechanism in several ways. By reducing the energy needed to operate, nodes operate more efficiently, reducing the overall cost and resource usage. It also reduces the costs of running a consensus mechanism, making it more accessible to a wider range of users and organizations. By reducing it, a consensus mechanism has a positive impact on the environment, making it more sustainable and socially responsible. The RBCET minimizes energy consumption by calculating the consensus score. If any node's consensus score is greater than the θ then it can participate in the consensus process, otherwise, it cannot. Fig. 9 shows that the simulation result of RBCET produces lower energy consumption than PBFT [10], [11], PoS [6], [7], and PoA [8], [9] by 52%, 14%, and 37% respectively.

D. Pending Transactions per Block

We have evaluated the efficiency of pending transactions per block across different consensus mechanisms like PBFT, PoS, PoA, and RBCET in Fig. 10. It refers to the set of unconfirmed transactions that the consensus mechanism is awaiting to include in the following block.

PBFT struggled with efficiency, showing an average of 280 pending transactions per block. This points to challenges in handling varying transaction volumes effectively.

PoS demonstrated better efficiency but still had an average of 70 pending transactions per block, indicating potential congestion during high-demand periods.

PoA performed moderately well with an average of 155 pending transactions per block, but its reliance on a limited number of validators posed some efficiency concerns.

The proposed **RBCET** exhibited exceptional performance, maintaining an average of only 18 pending transactions per block. This significant reduction highlighted RBCET's efficiency and suitability for real-time energy trading scenarios.

The number of pending transactions is crucial for efficient processing and dynamic energy trading environments between nodes. The RBCET's ability to keep pending transactions to a minimum signifies its potential to revolutionize energy trading on the blockchain, offering a more responsive and agile trading ecosystem. Its outstanding performance in managing pending transactions per block positions it as a promising consensus mechanism for energy trading, outshining PBFT [10], [11], PoS [6], [7], and PoA [8], [9] in terms of efficiency and responsiveness.

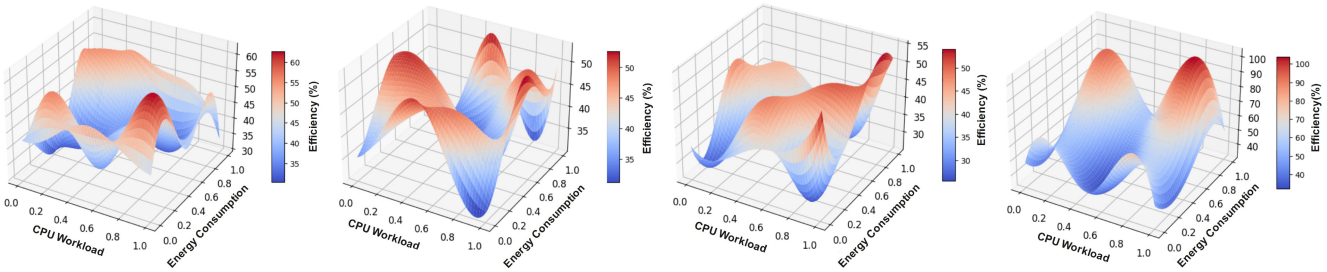
TABLE V
EFFICIENCY COMPARISONS OF RBCET WITH STATE-OF-THE-ART SCHEMES

Schemes	α	β	γ	δ	ϵ	Efficiency
PBFT [10], [11]	0.6	0.8	0.7	0.5	0.4	55%
PoS [6], [7]	0.6	0.6	0.6	0.4	0.4	50%
PoA [8], [9]	0.5	0.7	0.8	0.6	0.5	52%
RBCET	0.4	0.5	0.5	0.2	0.3	98%

E. Efficiency

The efficiency assessment of blockchain consensus mechanisms for energy trading between nodes entails a holistic approach considering many dimensions, such as computational efficiency, energy conservation, bandwidth utilization, and security robustness. To quantitatively capture these intricate inter-dependencies, we introduce a multidimensional efficiency formula meticulously tailored to the nuances of each consensus mechanism. The genesis of our efficiency formula lies in recognizing the complex trade-offs inherent in energy trading ecosystems. We began by distilling the critical factors impacting efficiency: CPU workload, energy consumption, bandwidth utilization, and security considerations. Acknowledging the intricate interplay among these dimensions, we formulated the comprehensive efficiency as $Efficiency = \alpha \cdot W_{CPU}^{\beta} \cdot E_{Energy}^{\gamma} \cdot U_{Bandwidth}^{\delta} \cdot S_{Security}^{\epsilon}$. Here, CPU Weight ($\alpha$), CPU Exponent ($\beta$), Energy Exponent($\gamma$), Bandwidth Exponent ($\delta$), and Security Exponent(ϵ) are coefficients that encapsulate the relative importance of each dimension. These are determined through an extensive optimization process. W_{CPU}, E_{Energy}, $U_{Bandwidth}$, and $S_{Security}$ are the normalized values of CPU workload, energy consumption, bandwidth utilization, and security respectively. The determination of the coefficients α, β, γ, δ, and ϵ for

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.



(a) PBFT [10], [11] (b) PoS [6], [7] (c) PoA [8], [9] (d) RBCET

Fig. 11. Efficiency values.

each consensus mechanism plays a pivotal role in quantifying their efficiencies within the context of energy trading for electric vehicles. As shown in Fig. 11(a), 11(b), 11(c), 11(d), and Table V, these coefficients are products of a rigorous optimization process conducted on our experimental testbeds, aligning with the specified maximum efficiency values for each mechanism.

VII. Conclusion and Future Work

The proposed scheme aims to enhance the security and efficiency of energy trading in the blockchain-based IoEV network by introducing the RBCET consensus mechanism. It leverages a reputation-based system that evaluates nodes using a DTR model and ensures that only nodes with an RS value exceeding the network average participate in energy trading. It mitigates risks such as double-spending, DDoS, eclipse, Sybil, and collusion attacks in the blockchain-based IoEV network. Implementing a hardware-based incentive mechanism enhances the efficiency of energy trading by incentivizing nodes to maintain high standards of participation and contribution. Our developed testbed, utilizing ten Raspberry Pi 4 devices, demonstrates significant enhancements over state-of-the-art consensus mechanisms. It shows a reduction in CPU workload by 47%, energy consumption by 10%, bandwidth usage by 5%, and volume of pending transactions by 75% which improves an overall efficiency improvement by 62%.

In future work, we aim to enhance the privacy of the nodes for energy trading in blockchain-based IoEV networks by using enhanced zero-knowledge proof.

References

[1] S. Zhang, S. Zhang, L. K. Yeung, and J. J. Q. Yu, "Urban Internet of Electric Vehicle parking system for vehicle-to-grid scheduling: Formulation and distributed algorithm," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 67–79, Jan. 2024.

[2] Bhawana et al., "BEET: Blockchain enabled energy trading for E-mobility oriented electric vehicles," *IEEE Trans. Mobile Comput.*, vol. 23, no. 4, pp. 3018–3034, Apr. 2024.

[3] A. Shahidinejad and J. Abawajy, "Blockchain-based self-certified key exchange protocol for hybrid electric vehicles," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 543–553, Feb. 2024.

[4] G. Sun, M. Dai, F. Zhang, H. Yu, X. Du, and M. Guizani, "Blockchain-enhanced high-confidence energy sharing in Internet of Electric Vehicles," *IEEE Internet Things J.*, vol. 7, no. 9, pp. 7868–7882, Sep. 2020.

[5] J. Kang, Z. Xiong, D. Niyato, D. Ye, D. I. Kim, and J. Zhao, "Toward secure blockchain-enabled Internet of Vehicles: Optimizing consensus management using reputation and contract theory," *IEEE Trans. Veh. Technol.*, vol. 68, no. 3, pp. 2906–2920, Mar. 2019.

[6] S. King and S. Nadal. "Ppcoin: Peer-to-peer crypto-currency with proof-of-stake." 2012. [Online]. Available: https://decred.org/research/king2012.pdf

[7] Q. Kong, R. Lu, F. Yin, and S. Cui, "Blockchain-based privacy-preserving driver monitoring for MaaS in the vehicular IoT," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3788–3799, Apr. 2021.

[8] P. Ekparinya, V. Gramoli, and G. Jourjon, "The attack of the clones against proof-of-authority," 2019, *arXiv:1902.10244*.

[9] R. Khalid, M. W. Malik, T. A. Alghamdi, and N. Javaid, "A consortium blockchain based energy trading scheme for electric vehicles in smart cities," *J. Inf. Security Appl.*, vol. 63, Dec. 2021, Art. no. 102998.

[10] M. Castro and B. Liskov, "Practical Byzantine fault tolerance," in *Proc. OsDI*, vol. 99, 1999, pp. 173–186.

[11] J. Mišić, V. B. Mišić, and X. Chang, "Design of proof-of-stake PBFT algorithm for IoT environments," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 2497–2510, Feb. 2023.

[12] J. Xu, C. Wang, and X. Jia, "A survey of blockchain consensus protocols," *ACM Comput. Surv.*, vol. 55, no. 13, pp. 1–35, 2023.

[13] T. Gresham. "Honda, GM-led group develops global blockchain standard for EV grid integration." 2020. [Online]. Available: https://shorturl.at/kyr6x

[14] C. Liu, K. K. Chai, X. Zhang, and Y. Chen, "Proof-of-benefit: A blockchain-enabled EV charging scheme," in *Proc. IEEE 89th Veh. Technol. Conf.*, 2019, pp. 1–6.

[15] Z. Su, Y. Wang, Q. Xu, M. Fei, Y.-C. Tian, and N. Zhang, "A secure charging scheme for electric vehicles with smart communities in energy blockchain," *IEEE Internet Things J.*, vol. 6, no. 3, pp. 4601–4613, Jun. 2019.

[16] Y. Wang et al., "A fast and secured vehicle-to-vehicle energy trading based on blockchain consensus in the Internet of Electric Vehicles," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 7827–7843, Jun. 2023.

[17] H. N. Abishu, A. M. Seid, Y. H. Yacob, T. Ayall, G. Sun, and G. Liu, "Consensus mechanism for blockchain-enabled vehicle-to-vehicle energy trading in the Internet of Electric Vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 1, pp. 946–960, Jan. 2022.

[18] A. Barnawi, S. Aggarwal, N. Kumar, D. M. Alghazzawi, B. Alzahrani, and M. Boulares, "Path planning for energy management of smart maritime electric vehicles: A blockchain-based solution," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 2, pp. 2282–2295, Feb. 2023.

[19] D. Kaur, G. S. Aujla, N. Kumar, A. Y. Zomaya, C. Perera, and R. Ranjan, "Tensor-based big data management scheme for dimensionality reduction problem in smart grid systems: SDN perspective," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 10, pp. 1985–1998, Oct. 2018.

[20] S. Ong and N. Clark, "Commercial and residential hourly load profiles for all TMY3 locations in the united states," DOE Open Energy Data Initiative, Nat. Renew. Energy Lab., Golden, CO, USA, Rep. 153, 2014.

[21] M. U. Javed et al., "Blockchain based secure, efficient and coordinated energy trading and data sharing between electric vehicles," *Clust. Comput.*, vol. 25, pp. 1–29, Jun. 2022.

Authorized licensed use limited to: University of Melbourne. Downloaded on November 03,2026 at 12:45:24 UTC from IEEE Xplore. Restrictions apply.